IMPACT OF VEHICLE ATTRIBUTES ON THE RESALE PRICES OF USED TOYOTA COROLLA CARS

by
H.P.D. Poornima
SC/2022/12951

Supervisor: Dr. K.D.Prasangika

Project report submitted to the
Department of Mathematics
in partial fulfillment of the requirement for the
Level II Case study course unit
of the

Bachelor of Science (Honors) Financial Mathematics and Industrial statistics

in the
Faculty of Science
University of Ruhuna
Matara
Sri Lanka.

August 2024

DECLARATION

I, H.P.D. Poornima , declare that the presented project report titled, "IMPACT OF VEHICLE ATTRIBUTES ON THE RESALE PRICES OF USED TOYOTA COROLLA CARS" is uniquely prepared by me based on the group project carried out under the supervision of Dr. K.D.Prasangika Department of Mathematics, Faculty of Science, University of Ruhuna, as a partial fulfillment of the requirements of the level II Case study course unit of the Bachelor of Science (Honors) Financial Mathematics and Industrial statistics in Department of Mathematics, Faculty of Science, University of Ruhuna, Sri Lanka. It has not been submitted to any other institution or study program by me for any other purpose.

Signature:.....

Date:.....

SUPERVISOR'S RECOMMENDATION

I/We certify that this study was carried out by H.P.D. Poornima under my/our supervision.

Signature:.....

Date:.....

Dr. K.D.Prasangika,

Department of Mathematics,

Faculty of Science,

University of Ruhuna.

ACKNOWLEDGMENTS

I would like to extend my sincere gratitude to Dr. for his exceptional mentorship, unwavering support, and invaluable guidance throughout this case study. Dr. K.D.Prasangika's expertise, insight, and encouragement have been instrumental in shaping the direction and outcomes of this study. Her dedication to excellence and commitment to fostering academic growth have inspired and motivated me to achieve my best. we are profoundly grateful for her mentorship. I would also like to express my gratitude to Mrs. Ruchira Kulasekara for her valuable contri butions and support during this research. Mrs Richira's advice and guidelines have been appreciated and have contributed to the overall success of this case study. Also, our sincere gratitude goes to the Department of Mathematics lecturers, for their support and guidance in keeping us progressing in conducting this case study. We would like to extend our thanks to our colleagues, who provided valuable feedback and engaged in helpful discussions, enhancing the case study process. Our special thanks goes to every individual who is involved in this case study for their contributions and support throughout the journey.

ABSTRACT

This study aims to predict the resale price of used cars using a Multiple Linear Regression (MLR) approach. A dataset containing 248 records, including data for Toyota Corolla vehicles, was analyzed. The dataset features variables such as Price, Age, Mileage, Color, Engin capacity and weight. Initial data exploration includes summary statistics, box plots and correlation matrices to assess variable distribution and relationships. These findings will provide useful insights for both buyers and sellers in determining fair market values.

Keywords: Used Car Price Prediction, Multiple Linear Regression, Toyota Corolla, Vehicle Valuation, Resale Value Estimation.

TABLE OF CONTENTS

TABLE OF CONTENTS iv

LIST OF TABLES vi

LIST OF FIGURES vii

1 Introduction 1

1.1 Background of the study 1
1.2 Problem statement 1
1.3 Significance of the study 1
1.4 Aims and Objectives 2
1.5 Research Question 2
1.6 Hypothesis 2

2 Literature Review 3

3 Data 5

3.1 About the Dataset 5
3.2 Data Dictionary 5
3.2.1 Dummy Variables for Property Style 6
3.3 Data Set Preparation 8

4 Method and Materials 9

4.1 Research approach 9
4.2 Conceptual model 10
4.3 Research Design 11

5 Results 12

5.1 Exploratory data Analysis 12
5.2 Quantitative Data Analysis 19
5.2.1 Correlation Analysis 19
5.2.2 Regression Analysis 20
5.2.3 Estimate model Parameters 23
5.2.4 Hypothesis Testing 25

6 Discussion and Conclusion 32

6.1 Discussion 32
6.2 Conclusion 33

References 34

A 35

A.1 R code for Exploratory Analysis 35

B R codes 37

B.1 R code for the multiple linear regression model 37

LIST OF TABLES

3.1 Dummy variable codes for Color 7

3.2 Dummy variable codes for Engine size 8

LIST OF FIGURES

5.1 Descriptive statistics table 12

5.2 Box plot of Price 12

5.3 Histogram of Price 12

5.4 Box plot of Age 13

5.5 Histogram of Age 13

5.6 scatter plot of Age. 13

5.7 Box plot of KM 14

5.8 Histogram of KM 14

5.9 scatter plot of KM 14

5.10 Box plot of Weight 15

5.11 Histogram of Weight 15

5.12 scatter plot of Weight 15

5.13 pie chart of color 16

5.14 Pie chart of CC 17

5.15 Corrplot of variables 18

5.16 Correlation Matrix 19

5.17 Box plot of Price 20

5.18 Histogram of Price 20

5.19 Descriptive statistics table 20

5.20 VIF value of the full model 21

5.21 Homoscedasticity 22

5.22 Summary of the full model 23

5.23 ANOVA table of the full model 24

5.24 Summary of forward model 24

5.25 Summary of forward model 25

5.26 Summary of the reduced model 28

5.27 ANOVA table of reduced model 28

CHAPTER 1
Introduction

1.1 Background of the study

The used car market plays a key role in the automotive industry by offering more affordable option to consumers while enabling vehicle owners to recover some value from their previous purchase. Normally, the resale value of a vehicle is affected by various factors such as Age, Mileage, Brand, Model, Engine capacity and Color. This study focuses on applying Multiple Linear Regression technique to a dataset of used cars, including Toyota corolla (2001-2004) models to analyze how features such as Age, Mileage, Engine size, Color and Weight influence resale price.

1.2 Problem statement

In this study, we focus on identifying the key factors that affect the resale price of used cars, with a special focus on Toyota Corolla models. In recent years, the used car market has experienced significant growth due to economic challenges and rising new vehicle costs. However, the lack of systematic and transparent method for pricing used vehicles often leads to unfair valuations in the market. Therefore, this study aims to address those issues by applying Multiple Linear Regression to model and predict resale prices based on selected vehicle features. As well as this study is connected to search answer of the following questions related to Used Car price determinants:

• What is the pattern of resale price for Toyota corolla vehicles in the selected market?

• How do factors such as Age, Mileage, Engine size, Color and Weight affect the resale price?

1.3 Significance of the study

This study provides valuable findings into the factors affecting the resale price of used Toyota corolla cars in Sri Lankan Market. Both buyers and sellers can make fair and informed decisions.Additionally ,assist policy makers or dealers in developing data-driven pricing strategies specific to the Sri Lankan market.

1.4 Aims and Objectives

The main objective of the study is to identify relationship between car attributes and the price of the used cars.

- To identify the relationship between the car attributes and the price of the used car
- To identify the relationship between age and price of the used car
- To identify the relationship between millage and price of the used car
- To identify the relationship between weight and price of the used car
- identify the relationship between color and price of the used car
- To identify the relationship between engine size and price of the used car

1.5 Research Question

- What is the relationship between the car attributes and the price of the used car?
- What is the relationship between age and price of the used car?
- What is the relationship between millage and price of the used car?
- What is the relationship between weight and price of the used car?
- What is the relationship between color and price of the used car?
- What is the relationship between engine size and price of the used car?

1.6 Hypothesis

H0 : Null hypothesis

There is no linear relationship between dependent variable and independent variables.

H1 : Alternative hypothesis

There is a linear relationship between dependent variable and independent variables

CHAPTER 2
Literature Review

In the study of A Multiple Regression Analysis of Factors that affect the reselling Price of a Car (N. Weeraratne,2016), the researcher investigated the impact of several variables: age of the car, mileage, car model (Avalon, Camry, Corolla, Matrix, Yaris) and car color (light or dark) on the resale price of used Toyota cars in the Colombo urban area. The study applied multiple linear regression analysis using a sample of 55 cars collected through stratified random sampling. The results showed that all included variables were statistically significant with newer cars, light colored cars and premium models such as Avalon and Camry. The final model demonstrated a strong fit with an adjusted R^2 of 97.0%. (N. Weeraratne,2016). In the study of Car Price Prediction using Machine Learning Techniques (Y.S. Balcıoğlu, B.Sezen, 2024), the researchers investigated the use of Machine learning models to predict car price based on a wide range of variables such as Brand, Model, Car conditions, Year, Power, Mileage, Fuel type, Transmission, Color, Number of doors and several Boolean features (Navigation system, Parking sensors and Heated seats). The study employed multiple Machine Learning algorithms, including Random Forest (RF), Support Vector Machine (SVM) and Artificial Neutral Network (ANN). This ensemble approach was found to significantly improve predictive accuracy, achieving a notable accuracy rate of 92.38%. (Y.S. Balcıoğlu, B.Sezen, 2024) In the study of Used Cars Price Prediction and using Data Mining Techniques (AlShared,2021), the researcher aimed to predict used car prices in the UAE using data scraped from local car selling websites with several variables such as Brand, Model, Production Year, Body Type, Color, Gear Type, Engine Size, Mileage, Specs Origin, Fuel Type and Price. The study applied three regression models: Random Forest Regressor, Linear Regression and Bagging Regressor. The model evaluation was conducted using metrics such as R^2, MSE, MAE and RMSE. Among the models, Random Forest Regressor achieved higher performance than the other models, achieving 95% accuracy, followed by Bagging Regressor (88%) and Linear regression (85%). The results indicated that Random Forest provided the most accurate prediction due to its ability to handle complex, non-linear data structures.

(AlShared,2021) In the study of Probabilistic Tabular Regression for used Car Pricing (Madhusudhanan et al.,2024), the researcher introduced a deep learning model called ProbSAINT, designed to enhance used car price prediction by incorporating uncertainty quantification. The model used a large database of 2 million records with features such as Brand, Model, Mileage, Engine size, Condition, Transmission, Fuel type and Offer duration. ProbSAINT was trained to predict both the price and its uncertainty, using a special loss function based on probability (log-likelihood). The model gave the most reliable prediction, especially when the model was highly confident, making it useful for real-world car pricing tasks. (Madhusudhanan et al.,2024) Research Gap While previous studies have explored used car prediction using advanced Machine learning techniques such as Random Forests and deep learning models. Apart from that many studies incorporate a wide variety of vehicles types and international markets. There remains a gap in research focusing on simpler, interpretable model like Multiple Linear Regression applied to a focused dataset within a specific regional context.

CHAPTER 3
Data

3.1 About the Dataset

The dataset is used to analyze the factors that affect the price of second-hand cars. It consists of 248 records and 6 variables. The dependent variable is the second-hand car price, and the independent variables include both numerical and categorical types.

The source of the dataset is Kaggle:

`https://www.kaggle.com/datasets/vishakhdapat/price-of-used-toyota-corolla-cars`

3.2 Data Dictionary

Variable Descriptions

1. **Price**: Represents the selling price of a used Toyota Corolla car. It is the dependent variable.
2. **Age**: Indicates the age of the car in years.
3. **KM**: The number of kilometers the car has traveled.
4. **Color**: The color of the car.This is a categorical Variable
5. **CC**: The engine size in cubic centimeters, indicating the total volume of all the engine's cylinders.CC is a categorical variable.
6. **Weight**: The kerb weight of the car, which includes standard equipment, fluids, and a full fuel tank, but excludes passengers and cargo.

3.2.1 Dummy Variables for Property Style

When converting the categorical variable Property Style into dummy variables, we use the following steps:

Color

- **Identify Categories:** The Property Style column has seven categories:
  - Black
  - Blue
  - Green
  - Grey
  - Red
  - Silver
  - Yellow
- **Choose a Reference Category:** We select Black as the reference category. This means we will not create a dummy variable for Black, and it will be represented when other dummy variables are 0.
- **Create Dummy Variables:** Define six dummy variables for the remaining categories:
  - Blue: 1 if the property is a Blue, 0 otherwise.
  - Green: 1 if the property is a Green, 0 otherwise.
  - Grey: 1 if the property is a Grey, 0 otherwise.
  - Red: 1 if the property is a Red, 0 otherwise.
  - Silver: 1 if the property is a Silver, 0 otherwise.
  - Yellow: 1 if the property is a Yellow, 0 otherwise.

Thus, the dummy variable coding will be as follows:

Property Style	Black	Blue	Green	Grey	Red	Silver	Yellow
Black	1	0	0	0	0	0	0
Blue	0	1	0	0	0	0	0
Green	0	0	1	0	0	0	0
Grey	0	0	0	1	0	0	0
Red	0	0	0	0	1	0	0
Silver	0	0	0	0	0	1	0
Yellow	0	0	0	0	0	0	1

Table 3.1: Dummy variable codes for Color

Engine size

- **Identify Categories:** The Property Style column has four categories:
  - CC1398
  - CC1400
  - CC1598
  - CC1600
- **Choose a Reference Category:** We select CC1398 as the reference category. This means we will not create a dummy variable for CC1398, and it will be represented when other dummy variables are 0.
- **Create Dummy Variables:** Define three dummy variables for the remaining categories:
  - CC1400: 1 if the property is a Blue, 0 otherwise.
  - CC1598: 1 if the property is a Green, 0 otherwise.
  - CC1600: 1 if the property is a Grey, 0 otherwise.

Thus, the dummy variable coding will be as follows:

Property Style	CC1398	CC1400	CC1598	CC1600
CC1398	1	0	0	0
CC1400	0	1	0	0
CC1598	0	0	1	0
CC1600	0	0	0	1

Table 3.2: Dummy variable codes for Engine size

3.3 Data Set Preparation

Check for Null Values

We checked whether the dataset contains any null values using the following R code:

```
> is.null(data)
```

The result showed that the dataset does not contain any null values.

Check for NA Values

We also verified whether the dataset contains any `NA` (Not Available) values:

```
> is.na(data)
```

The result indicated that the dataset does not contain any NA values either.

Outliers

Some extreme outliers were identified and removed during the data preparation process to reduce the effect on the regression results.

Data Transformation

A new variable `Age` was created by combining the `Manufacture Year` and `Manufacture Month` to improve the linear relationship with the dependent variable.

CHAPTER 4
Method and Materials

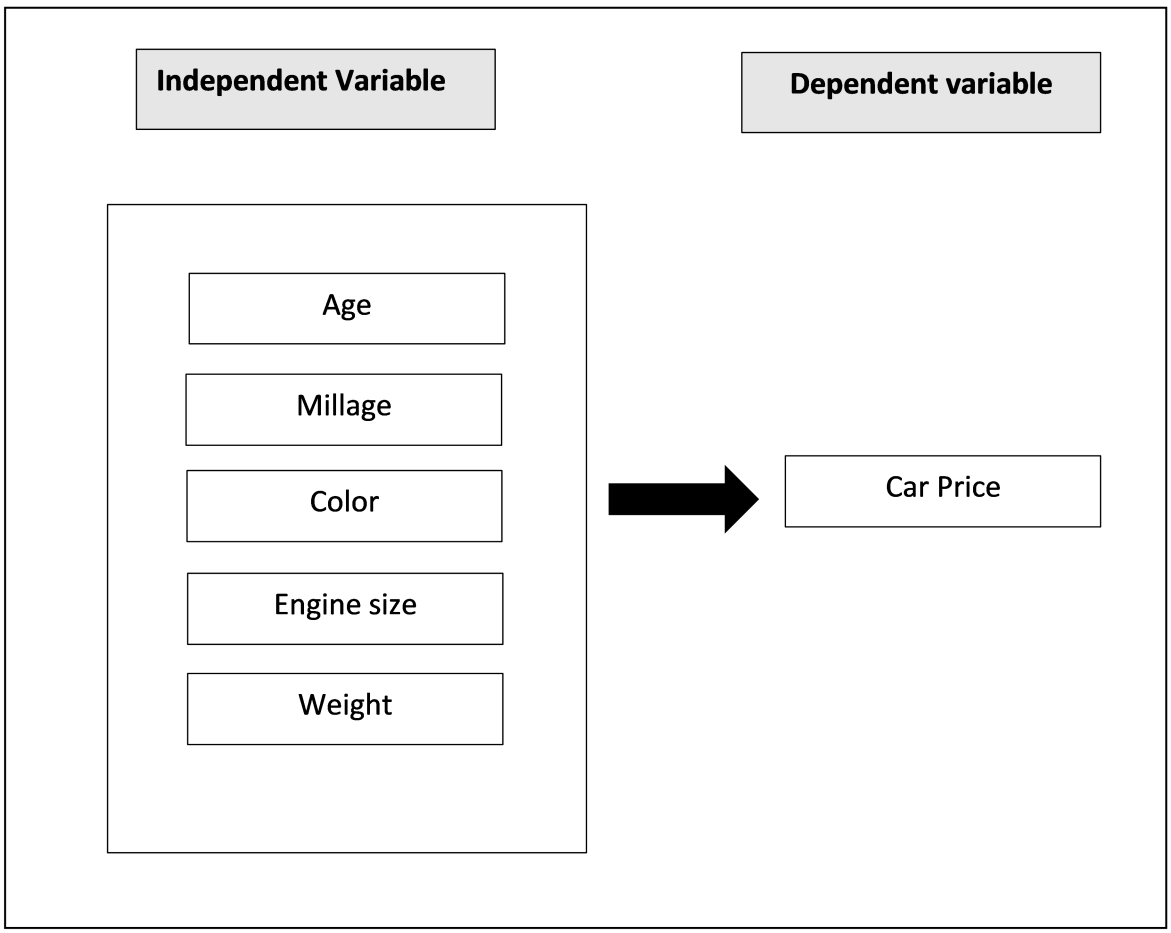
4.1 Research approach

The overall research approach for this study is qualitative research approach.

Qualitative research approach

Quantitative research is a method of collecting and analyzing numerical data to identify patterns, test hypothesis and make predictions. The multiple linear regression model is one of the tools of quantitative approach. It is used to determine a mathematical relationship among several variables. In our case, We have several independent variables and we need to identify the relationship between variables. Therefore, we selected multiple linear regression as our model.

4.2 Conceptual model



4.3 Research Design

The main purpose of the study is to use multiple linear regression to identify how second hand manufacture year, KM Driven ,color, Engine capacity, weight and quarterly tax impact on the second hand car price .The step wise approach was adopted due to the fact that it allows for the recognition of the most relevant independent variables. After that using the most relevant independent variables, the best-fitted model is constructed for making conclusion.

Multiple linear regression is a statistical technique that uses several independent variables to predict the outcome of a response variable. A population model for multiple linear regression is written as

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \varepsilon \tag{4.1}$$

Where:

- Y = Dependent variable (Resale price of used car)
- $X_1, X_2, \ldots, X_k$ = Independent variables (e.g.Age, KM driven, color, engine size, weight)
- β_0 = Intercept
- $\beta_1, \beta_2, \ldots, \beta_k$ = Regression coefficients for the respective independent variables
- ε = Error term

CHAPTER 5
Results

5.1 Exploratory data Analysis

Descriptive Statistics: Price, Age, KM, CC, Weight

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Price	248	0	14660	177	2785	10950	12413	13625	16839	20950
Age	248	0	21.901	0.0558	0.878	19.833	21.250	22.167	22.583	22.917
KM	248	0	34694	1109	17469	450	22899	32875	45607	96829
CC	248	0	1533.0	6.00	94.6	1398.0	1400.0	1600.0	1600.0	1600.0
Weight	248	0	1085.1	2.11	33.2	1025.0	1060.0	1080.0	1110.0	1180.0

Figure 5.1: Descriptive statistics table

Price

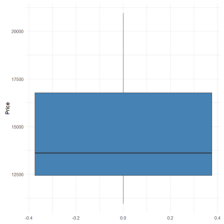


Figure 5.2: Box plot of Price

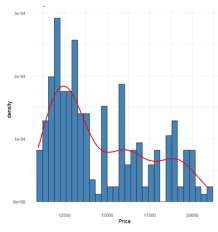


Figure 5.3: Histogram of Price

The histogram of Price shows a distribution that is not perfectly normal. There is no symmetrical bell shape. The boxplot of Price indicates a relatively symmetrical distribution. In conclusion, the 'Price' variable's distribution is not perfectly normal and appears to be slightly positively skewed.

Age

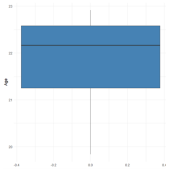


Figure 5.4: Box plot of Age

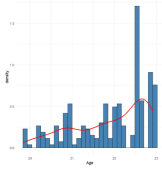


Figure 5.5: Histogram of Age

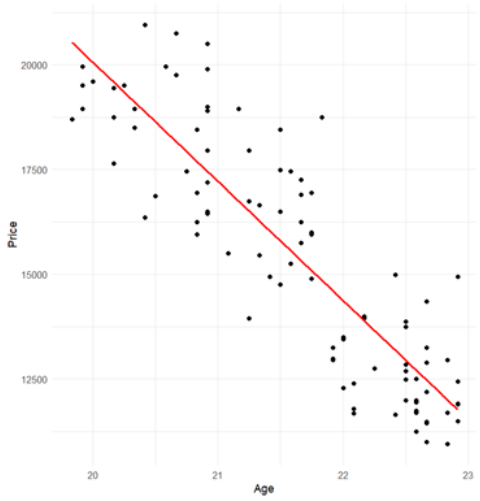


Figure 5.6: scatter plot of Age.

The histogram shows a distribution that is clearly not perfectly normal. It appears to be multimodal and has a strong left skewness . The boxplot of Age reinforces this left skewness. This also confirms that the distribution is not perfectly normal. The scatterplot shows a strong negative linear relationship between the two variables, suggesting good linearity. the 'Age' variable is left-skewed and not normally distributed, Therefore,'Age' shows a strong negative linear relationship with 'Price'.

Millage(KM)

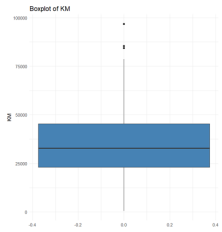


Figure 5.7: Box plot of KM

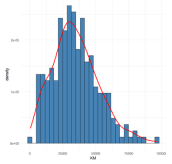


Figure 5.8: Histogram of KM

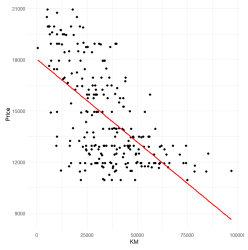


Figure 5.9: scatter plot of KM

The histogram shows a positively skewed and clearly indicates the distribution is not perfectly normal. The boxplot also confirms the positive skewness. Both plots show that the 'KM' variable has a wide spread. The scatterplot demonstrates a negative linear relationship and a clear downward trend line can be observed, suggesting good linearity. Overall, the 'KM' variable is positively skewed and not normally distributed.overall the KM shows a strong negative linear relationship with 'Price'.

Weight

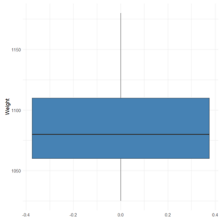


Figure 5.10: Box plot of Weight

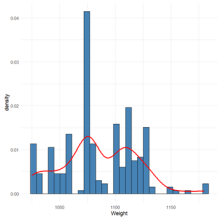


Figure 5.11: Histogram of Weight

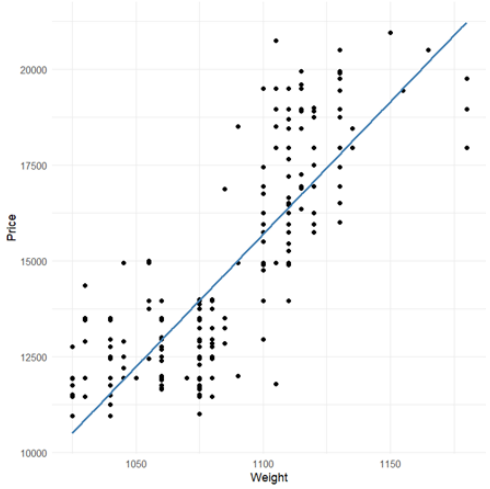


Figure 5.12: scatter plot of Weight

The histogram shows a positively skewed distribution, longer tail extending towards higher weight values. This indicates the distribution is not perfectly normal. The boxplot also confirms the positive skewness. This also supports that the distribution is not perfectly normal. The scatterplot shows a positive linear relationship between the price and weight and a clear upward trend line can be observed, suggesting good linearity. According to the plots, 'Weight' variable is positively skewed and not normally distributed.

Color

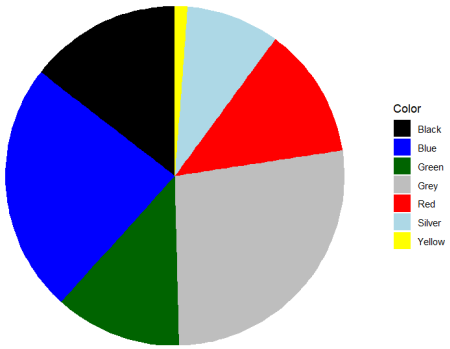


Figure 5.13: pie chart of color

The pie chart shows the color of the cars and the highest percentage gives the gray-colored cars.

Engine size

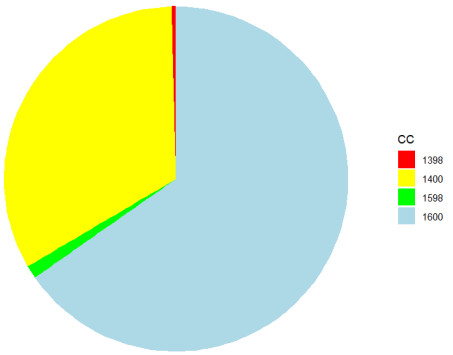


Figure 5.14: Pie chart of CC

The pie chart reveals the engine capacity of the Toyota Corolla cars. Most cars have a 1400 engine capacity.

Corr plot of variables

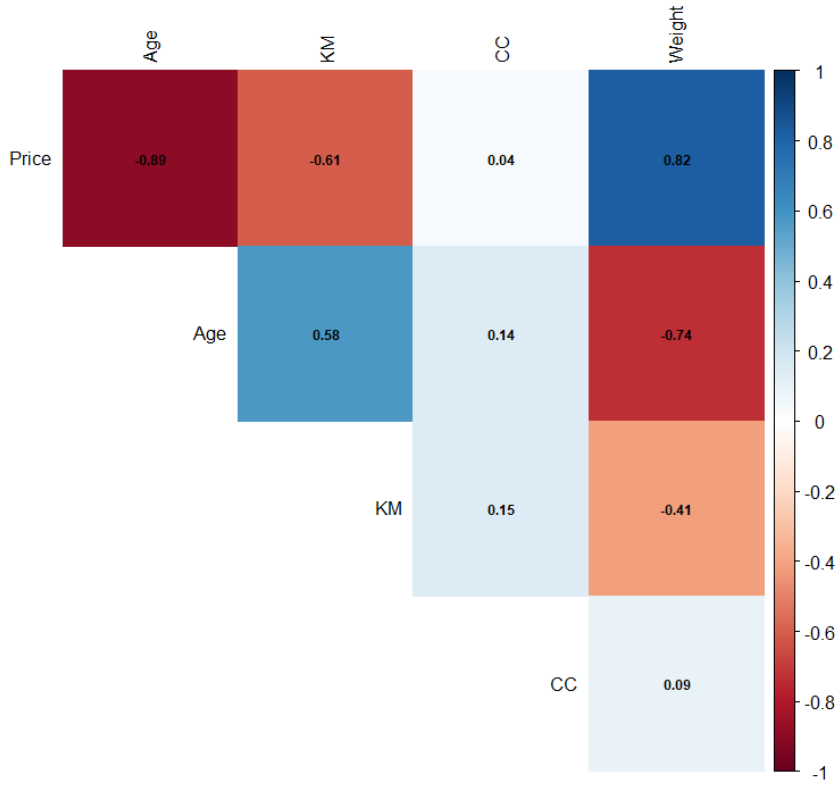


Figure 5.15: Corrplot of variables

The correlation plot reveals several interesting relationships between the variables. Price has strong negative relationship with Age exhibiting a particularly strong inverse connection and moderately strong negative relationship with KM. There is a positive relationship between Price and Weight, suggesting that heavier items are generally more expensive. Furthermore, CC demonstrates a weak relationship with price.

5.2 Quantitative Data Analysis

As mentioned earlier, there are no missing values and null values in our data set. In this quantitative data analysis part, we are trying to get the most suitable model for our data set and then we will get the idea about what factors are mainly affected to resale car price of Toyota corolla.

5.2.1 Correlation Analysis

A correlation table is a matrix that displays the correlation coefficients between multiple variables in a dataset. A positive correlation (closer to 1) indicates that as one variable increases, the other tends to also increase. A negative correlation (closer to-1) indicates that as one variable increases, the other tends to decrease. A correlation close to 0 indicates a weak or no linear relationship. Overall, the correlation table suggests that there are significant positive, negative, and no correlations between all these variables; between the dependent variable(Price) and all the other independent variables. Therefore, we can make decisions and understand the linear relationship of these variables.

	Price	Age	KM	CC	Weight
Price	1.0000	-0.8945	-0.6085	0.0367	0.8227
Age	-0.8945	1.0000	0.5783	0.1433	-0.7395
KM	-0.6085	0.5783	1.0000	0.1457	-0.4105
CC	0.0367	0.1433	0.1457	1.0000	0.0939
Weight	0.8227	-0.7395	-0.4105	0.0939	1.0000

Figure 5.16: Correlation Matrix

5.2.2 Regression Analysis

As mentioned in above there are no any duplicate values and null values. Then we have tried to get a fitted model for the dependent variable that we have and our data set maybe have some outliers at the moment when we are doing for the fit. As mentioned in below box plot of Price and histogram of price we can see there are no outliers in our data set. And also, if there are no any outliers of the data set.

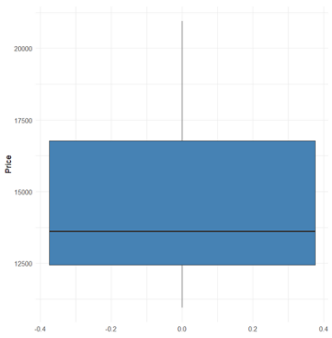


Figure 5.17: Box plot of Price

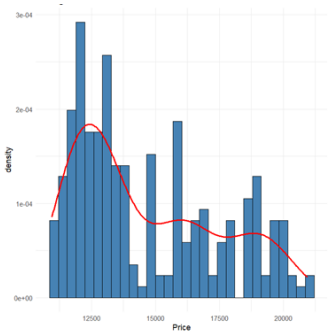


Figure 5.18: Histogram of Price

we can apply the regression to get the best model for our dataset.we can apply regression analyze if we can satisfy its assumptions.

Those assumptions are,

1. **The residual values are normally distributed.**

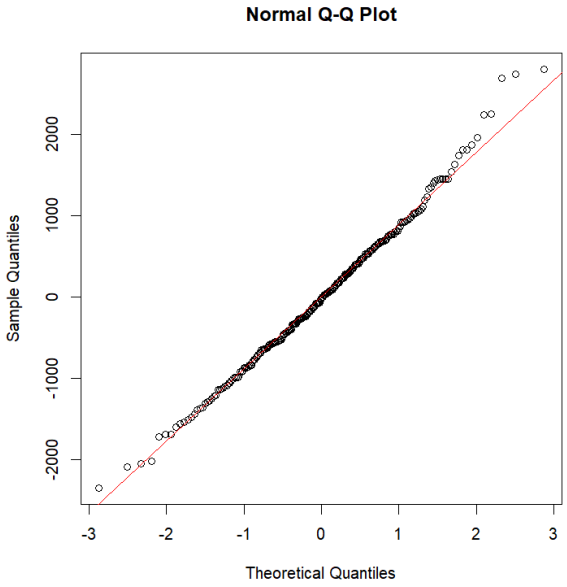


Figure 5.19: Descriptive statistics table

We can see 5.19is normally distributed. Then our this assumption is satisfied.

2. **There must be a linear relationship between the dependent and the independent variables.**

   As we have shown scatter plots in Exploratory Data Analysis part we can see all in dependent variables and the dependent variable have a linear relationship. As well as in the correlogram 5.15also we can see the relationship between independent and dependent variables. Then our this assumption is also satisfied.

3. **Multicollinearity**

   If there are no collinearity in the data then all the values should be less than 0.8 . This assumption is verified by computing a matrix of Pearson's correlation among all the independ variables. In the figure 5.16 (A matrix of Pearson's correlation) we can see that the independent variables are not highly correlated with each other. Then our this assumption is also satisfied.

   According to the below figure we also get VIF value of the model. In our data set, we have categorical the got GVIF values instead of VIF values. The GVIF results show that all predictors have adjusted GVIF values below table, indicating that multicollinearity is not an issue in the model. For all variables, the GVIF values are well within the acceptable range. Overall, the predictors are not strongly correlated with each other, ensuring stable regression estimates.

	GVIF	Df	GVIF^(1/(2*Df))
Age	3.087433	1	1.757109
KM	1.607628	1	1.267923
Color	1.322398	6	1.023560
CC	1.216964	1	1.103161
Weight	2.477688	1	1.574067

Figure 5.20: VIF value of the full model

4. **Homoscedasticity**

The homoscedasticity assumes that the variance of the residual error is similar across the value of each independent variable. We checked it by plotting predicted values against the standardized residual values.

Based on the scatterplot of predicted values vs. standardized residuals provided in Figure 5.21, we can make the following observations about homoscedasticity. The assumption of homoscedasticity appear to be satisfied in this case. According to the graph satisfies homoscedasticity property because there we can not see any tunnel shape data distribution.So the assumption is satisfied.

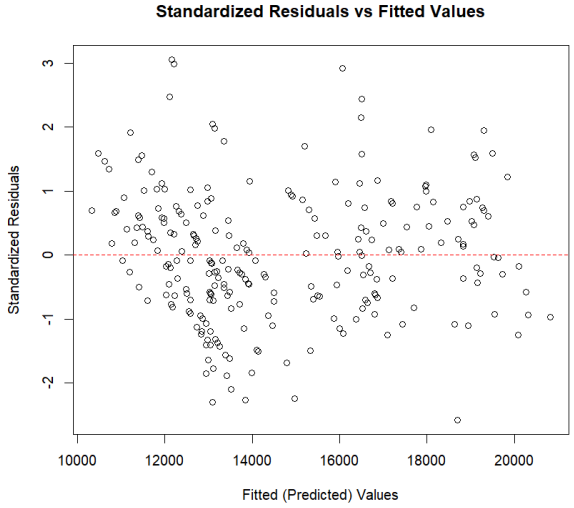


Figure 5.21: Homoscedasticity

5.2.3 Estimate model Parameters

In this section, relationship between the dependent and independent variables are mathematical represented. Then, which independent variables are to be included is chosen to assess the model's fit.

```
Call:
lm(formula = Price ~ Age + KM + Color + CC + Weight, data = df)

Coefficients:
(Intercept)          Age           KM    ColorBlue   ColorGreen    ColorGrey     ColorRed
 28211.6704   -1815.3994      -0.0266    -540.9494    -772.2289    -394.4731    -736.9984
ColorSilver  ColorYellow       CC1400       CC1598       CC1600       Weight
  -679.9128    -314.0227    -780.6952    1166.0995     -33.8375      25.6949

> summary(model)

Call:
lm(formula = Price ~ Age + KM + Color + CC + Weight, data = df)

Residuals:
     Min       1Q   Median       3Q      Max
-2348.74  -593.59   -24.73   603.12  2792.56

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.821e+04  5.180e+03   5.446 1.30e-07 ***
Age         -1.815e+03  1.192e+02 -15.235  < 2e-16 ***
KM          -2.661e-02  4.326e-03  -6.149 3.32e-09 ***
ColorBlue   -5.409e+02  1.982e+02  -2.730  0.00682 **
ColorGreen  -7.722e+02  2.381e+02  -3.243  0.00136 **
ColorGrey   -3.945e+02  1.984e+02  -1.988  0.04792 *
ColorRed    -7.370e+02  2.344e+02  -3.144  0.00188 **
ColorSilver -6.799e+02  2.589e+02  -2.626  0.00920 **
ColorYellow -3.140e+02  5.679e+02  -0.553  0.58080
CC1400      -7.807e+02  9.515e+02  -0.821  0.41276
CC1598       1.166e+03  1.097e+03   1.063  0.28891
CC1600      -3.384e+01  9.545e+02  -0.035  0.97175
Weight       2.569e+01  2.831e+00   9.076  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 934.4 on 235 degrees of freedom
Multiple R-squared:  0.8929,    Adjusted R-squared:  0.8875
F-statistic: 163.3 on 12 and 235 DF,  p-value: < 2.2e-16
```

Figure 5.22: Summary of the full model

According to figure 5.22 the multiple linear regression equation is

Price= 28210-1815 Age-0.02661 KM -540.9 ColorBlue-772.2 ColorGreen-394.5 ColorGrey -737 ColorRed- -679.9 ColorSilver-314 ColorYellow -780.7 CC1400+ 1166 CC1598 -33.84 CC1600 + 25.69 Weight

In the above figure 5.22 our full model R squared value is **89.29 %**

```
Analysis of Variance Table

Response: Price
           Df     Sum Sq    Mean Sq   F value    Pr(>F)
Age         1 1533283692 1533283692 1756.1247 < 2.2e-16 ***
KM          1   23945850   23945850   27.4260 3.623e-07 ***
Color       6   10408573    1734762    1.9869   0.06834 .
CC          3   71643817   23881272   27.3521 3.292e-15 ***
Weight      1   71915477   71915477   82.3674 < 2.2e-16 ***
Residuals 235  205180006     873106
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Figure 5.23: ANOVA table of the full model

Forward Elimination Method

This method starts with an empty model and adds variables one by one, testing each variable to see if its inclusion significantly improves the model fit based on a predefined criterion. It continues adding variables until no more variables meet the criterion for inclusion.There are 6 steps involved in the regression model's forward elimination procedure.The output of the forward elimination are shown below figures 5.24 and 5.25

```
                          Stepwise Summary
-------------------------------------------------------------------------------
Step    Variable        AIC         SBC         SBIC        R2       Adj. R2
-------------------------------------------------------------------------------
 0      Base Model    4641.141    4648.168    3934.000    0.00000    0.00000
 1      Age           4243.883    4254.423    3537.972    0.80009    0.79928
 2      Weight        4162.023    4176.077    3456.899    0.85745    0.85629
 3      KM            4137.914    4155.481    3433.137    0.87169    0.87012
 4      CC            4114.975    4143.083    3407.078    0.88583    0.88298
 5      Color         4111.034    4160.222    3393.898    0.89293    0.88747
-------------------------------------------------------------------------------

Final Model Output
------------------

                        Model Summary
-----------------------------------------------------------------
R                       0.945       RMSE                909.582
R-Squared               0.893       MSE              827338.735
Adj. R-Squared          0.887       Coef. Var             6.374
Pred R-Squared           -Inf       AIC                4111.034
MAE                   720.823       SBC                4160.222
-----------------------------------------------------------------
 RMSE: Root Mean Square Error
 MSE: Mean Square Error
 MAE: Mean Absolute Error
 AIC: Akaike Information Criteria
 SBC: Schwarz Bayesian Criteria

                                ANOVA
-------------------------------------------------------------------------------
                    Sum of
                   Squares         DF      Mean Square       F          Sig.
-------------------------------------------------------------------------------
Regression    1711197410.531        12    142599784.211    163.325    0.0000
Residual       205180006.368       235       873106.410
Total         1916377416.899       247
-------------------------------------------------------------------------------
```

Figure 5.24: Summary of forward model

Stepwise Summary

Step	Variable	AIC	SBC	SBIC	R2	Adj. R2
0	Base Model	4641.141	4648.168	3934.000	0.00000	0.00000
1	Age	4243.883	4254.423	3537.972	0.80009	0.79928
2	Weight	4162.023	4176.077	3456.899	0.85745	0.85629
3	KM	4137.914	4155.481	3433.137	0.87169	0.87012
4	CC	4114.975	4143.083	3407.078	0.88583	0.88298
5	Color	4111.034	4160.222	3393.898	0.89293	0.88747

Final Model Output

Model Summary

R	0.945	RMSE	909.582
R-Squared	0.893	MSE	827338.735
Adj. R-Squared	0.887	Coef. Var	6.374
Pred R-Squared	-Inf	AIC	4111.034
MAE	720.823	SBC	4160.222

RMSE: Root Mean Square Error
MSE: Mean Square Error
MAE: Mean Absolute Error
AIC: Akaike Information Criteria
SBC: Schwarz Bayesian Criteria

ANOVA

	Sum of Squares	DF	Mean Square	F	Sig.
Regression	1711197410.531	12	142599784.211	163.325	0.0000
Residual	205180006.368	235	873106.410		
Total	1916377416.899	247			

Figure 5.25: Summary of forward model

There no any variable add remove from the full model.so the forward Elimination method also gave the same output model.

5.2.4 Hypothesis Testing

- **Null hypothesis (H0):**

$$\beta(i) = 0, \quad i = 1, 2, 3, 4, 5 \tag{5.1}$$

There is no linear relationship between the dependent variable and the independent variables.

- **Alternative hypothesis (Ha):**

$$\beta(i) \neq 0, \quad i = 1, 2, 3, 4, 5 \tag{5.2}$$

Research question :Relationship between Age and Price

According to the summary of model we can see the P value is small than 0.05 for Age.Then We can reject H0.There is a linear relationship between Age and Price.

Research question :Relationship between KM and Price

According to the summary of the model we can see that the P value is smaller than 0.05 for KM.Then we can reject H0.There is a linear relationship between KM and Price.

Research question :Relationship between ColorBlue and Price

According to the summary of the model we can see that the P value is smaller than 0.05 for ColorBlue.Then we can reject H0.There is a linear relationship between ColorGreen and Price.

Research question:Relationship between ColorGrey and Price

According to the summary of the model we can see that the P value is smaller than 0.05 for KM.Then we can reject H0.There is a linear relationship between ColorGrey and Price.

Research question :Relationship between ColorRed and Price

According to the summary of the model we can see that the P value is smaller than 0.05 for KM.Then we can reject H0.There is a linear relationship between ColorRed and Price.

Research question :Relationship between Silver and Price

According to the summary of model we can see that the P value is smaller than 0.05 for KM.Then we can reject H0.There is a linear relationship between Silver and Price.

Research question:Relationship between ColorYellow and Price

According to the summary of model we can see the P value is bigger than 0.05 for KM.But we can't accept the Alternative hypothesis because ColorYellow is a dummy variable of the Color.

Research question:Relationship between CC1400 and Price

According to the summary of model we can see the P value is bigger than 0.05 for CC1400.Then We can't reject H0.There is a non-linear relationship between CC1400 and Price.

Research question:Relationship between CC1598 and Price

According to the summary of the model we can see that the P value is greater than 0.05 for KM.Then we can reject H0.There is a non-linear relationship between CC1598 and Price.

Research question:Relationship between CC1600 and Price

According to the summary of model we can see the P value is bigger than 0.05 for KM.Then We can't reject H0.There is a non-linear relationship between CC1600 and Price.

According to the Hypothesis of Research quesion 4,There is a no linear relationship between CC and price.

Research question:Relationship between Weight and Price

According to the summary of the model we can see the P value is smaller than 0.05 for Weight.Then We can reject H0.There is a linear relationship between Weight and Price.

Based on the result of full model and hypothesis we decided to remove the CC(Engine size) variable.Then we recreated the model and these are summary and ANOVA table of the reduced model.

```
Call:
lm(formula = Price ~ Age + KM + Color + Weight, data = df)

Residuals:
     Min       1Q   Median       3Q      Max
-2720.62  -668.27    -0.42   596.53  2996.60

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.944e+04  5.216e+03   3.727 0.000242 ***
Age         -1.664e+03  1.228e+02 -13.548  < 2e-16 ***
KM          -2.474e-02  4.597e-03  -5.381 1.77e-07 ***
ColorBlue   -4.834e+02  2.111e+02  -2.291 0.022866 *
ColorGreen  -6.570e+02  2.530e+02  -2.597 0.009994 **
ColorGrey   -4.721e+02  2.095e+02  -2.253 0.025160 *
ColorRed    -6.399e+02  2.493e+02  -2.567 0.010868 *
ColorSilver -4.843e+02  2.735e+02  -1.771 0.077878 .
ColorYellow -5.088e+02  6.043e+02  -0.842 0.400616
Weight       3.039e+01  2.858e+00  10.631  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 996.8 on 238 degrees of freedom
Multiple R-squared:  0.8766,    Adjusted R-squared:  0.8719
F-statistic: 187.9 on 9 and 238 DF,  p-value: < 2.2e-16
```

Figure 5.26: Summary of the reduced model

According to the figure 5.26 our reduced model R squared value is 87.66%

```
Analysis of Variance Table

Response: Price
           Df     Sum Sq    Mean Sq   F value    Pr(>F)
Age         1 1533283692 1533283692 1543.2648 < 2.2e-16 ***
KM          1   23945850   23945850   24.1017 1.698e-06 ***
Color       6   10408573    1734762    1.7461    0.1111
Weight      1  112278577  112278577  113.0095 < 2.2e-16 ***
Residuals 238  236460724     993532
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Figure 5.27: ANOVA table of reduced model

Hypothesis 2

- Null Hypothesis(H0)

  Reduced model is suitable

- Alternative Hypothesis(Ha)

  Full model is suitable

F Satistics

$$F = \frac{(SSE_{reduced} - SSE_{full})/(p-m)}{SSE_{full}/(n-p-1)}$$
$$= \frac{(236460724 - 205180006)/(12-9)}{(205180006)/(248-12-1)}$$

$$= 11.942$$

F Critical value, For 95% confidence Interval

F(1-0.05,3,235)= 2.64

$$Fpartial > Fcriticalvalue$$

Then we can reject null hypothesis. Therefore **Full model is suitable** at 5% significance level.

Interpretation of the full model

Price= 28210-1815 Age-0.02661 KM -540.9 ColorBlue-772.2 ColorGreen- 394.5 Color-Grey -737 ColorRed- -679.9 ColorSilver-314 ColorYellow -780.7 CC1400+ 1166 CC1598 -33.84 CC1600 + 25.69 Weight

- Intercept: when all the predictors are equal to zero, the Price is increased by 28210.

- Age (-1815): For one year change of age, the price of the car is predicted to decrease by 1815, assuming all other factors in constant.

- KM (-0.02661): For every kilometer the car has been driven, the price is predicted to decrease by 0.02661, assuming all other variables to be constant.

- Weight (+25.69): For one unit change of weight, the price of the car is predicted to increase by 25.69, assuming all other factors remain constant.

- ColorBlue (-540.9): A car with the color blue is predicted to have a price that is 540.9 lower than a car of the base color, all other factors being constant.

- ColorGreen (-772.2): A car with the color green is predicted to have a price that is 772.2 lower than a car of the base color.

- ColorGrey (-394.5): A car with the color gray is predicted to have a price that is 394.5 units lower than a car of the base color, all other factors being constant.

- ColorRed (-737): A car with the color red is predicted to have a price that is 737 lower than a car of the base color, all other factors being constant.

- ColorSilver (-679.9): A car with the color silver is predicted to have a price that is 679.9 lower than a car of the base color, all other factors being constant.

- ColorYellow (-314): A car with the color yellow is predicted to have a price that is 314 lower than a car of the base color, all other factors being constant.

- CC1400 (-780.7): A car with a Cubic Capacity of 1400 cc is predicted to have a price that is 780.7 lower than a car of the base CC, all other factors being constant..

- CC1598 (1166): A car with a Cubic Capacity of 1598 cc is predicted to have a price that is 1166 greater than a car of the base CC, all other factors being constant.

- CC1600 (-33.84): A car with a Cubic Capacity of 1600 cc is predicted to have a price that is 33.84 lower than a car of the base CC, all other factors being constant.

CHAPTER 6
Discussion and Conclusion

6.1 Discussion

This study used a Multiple linear regression approach to analyze a dataset of 248 of used Toyota Corolla Cars. The main aim of this study is predict resale prices based on attributes like Age, Millage(KM),Color ,Engine Capacity(CC) and Weight. First, we started descriptive analysis, get summary statics and draw plots. After we checked the normality, it was not a normally distributed. The during the data processing stage some extreme outliers were identified and removed to get the good correlation between the independents and dependent variable. In data Transformation stage we created a new variable Age using manufacturing year and month , which helped to .We have analyzed the data set using R software and we did correlation analysis, regression analysis. In regression analysis part we estimate models , model summaries and ANOVA table by using R software. First, we got the full model r squared value .it was 89.29 %. Then get best fitted model and r squared vale, we used Forward Elimination method. But we got the same method after using forward elimination method. After doing Hypothesis 1, We remove the CC (Engine size) variable from full model because the p values of the dummy variables of cc variable is smaller than 0.05 % significance level. Our reduced model R squared value is 87.19 % and When we doing hypothesis, we can answer the research question.

.Then we checked Hypothesis 2, in here we could know full reduced is more suitable than reduced model.Based on the statistical output,full is the better than reduced model. Full model's AIC and SBC values are 4111.034, 4160.22 .reduced mode has 4140.224 AIC value and 4178.871 Value. Full model AIC and SBC values smaller than reduced model and full model has a lower Residual Standard Error, a higher Adjusted R-squared, and its predictors are more statistically significant, making it a more accurate.Therefore we finally concluded that **Full model is most suitable model**.

6.2 Conclusion

This study set out to determine the impact of specific variables on resale price of used Toyota corolla cars. Based on the analysis the resale price is significantly and negatively affect by its Age and Millage (KM). The weight of the vehicle has a significant positive impact on resale price. This study found that every unit increase weight, that car's price is predicated to increase. The color of the vehicle is a significant of price, with certain colors negatively impacting the value compared to a baseline neutral color.CC(Engine Capacity) is also moderately affect to the Price . Based on the statistical output, the full model was determined to be more suitable than the reduced model. Finally, the study concluded that the full model, which includes the variables Age, Millage (KM), Color, Engine Capacity (CC), and Weight, is the most accurate for predicting the resale price of used Toyota Corolla cars.

References

[1] M. AlShared. Used cars price prediction using data mining techniques. international journal of data science and analytics,. *Statistics for Systems Biology Group. Research Report*, (7), 2007.

[2] Sezen B. Balcıo˘glu, Y. S. Car price prediction using machine learning techniques. journal of artificial intelligence research and applications, 12(1), 45–60. 2024.

[3] et al. Madhusudhanan, A. Probabilistic tabular regression for used car pricing: The probsaint model. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024.

[4] N. Weeraratne. A multiple regression analysis of factors that affect the reselling price of a car[unpublished study], 2016.

APPENDIX A

Dataset

You can find the final dataset for this analysis at the following link: Click here to open the dataset

A.1 R code for Exploratory Analysis

```r
# Load necessary libraries
library(readr)
library(dplyr)
library(corrr)
library(car)
library(corrplot)

# Load the data set
data <- read.csv("new_data_set4.csv")
numeric_data <- data %>% select(where(is.numeric))
numeric_data

#Regression Line and plot
# Plot Age vs Price with regression line
plot(df$Age, df$Price,
     main = "Age vs Price",
     xlab = "Age", ylab = "Price", col = "blue", pch = 16)
abline(lm(Price~Age, data = df), col = "red", lwd = 2)

#histogram

hist(df$Price,
     main = "Histogram of Price",
     xlab = "Price",
     col = "skyblue",
     border = "white")
#box plot
boxplot(df$Price,
```

```
        main = "box of Price",
        xlab = "Price",
        col = "skyblue")



# Calculate correlation matrix with respect to Price
cor_matrix <- cor(numeric_data, use = "complete.obs")

cor_with_target <- cor_matrix[, "Price"]
# Filter variables with high correlation (adjust threshold as needed)
high_cor_vars <- names(cor_with_target[abs(cor_with_target) > 0])
# Create a correlation matrix of only those high-correlation variables
final_cor_matrix <- cor_matrix[high_cor_vars,high_cor_vars ]
# Print the result
print(round(final_cor_matrix, 2))

corrplot(final_cor_matrix,
         method = "color", # Use colored squares
         type = "upper", # Show upper triangle only
         addCoef.col = "black", # Add coefficient text in black
         tl.col = "black", # Text label color
         number.cex = 0.6, # Size of the number labels
         tl.cex = 0.8, # Size of variable labels
         diag = FALSE)
```

APPENDIX B
R codes

B.1 R code for the multiple linear regression model

```
library(car)
library(dplyr)
library(knitr)
library(olsrr)
data<-read.csv(file="new_data_set4.csv")
df<-data.frame(data)
head(df)

df$Color <- as.factor(df$Color  )
df$CC<- as.factor(df$CC )
model <- lm(Price~Age+KM+Color+CC+Weight, data = df)
summary(model)




numeric_data <- df %>%
  select_if(is.numeric)
correlation_matrix <- round(cor(numeric_data),4)

#cor matrix table
kable(correlation_matrix,
      caption = "Correlation Table of Numeric Variables")
summary_data <- summary(df)
summary_data
#residual plot
plot(model, which = 1)

# Q-Q plot for normality
qqnorm(residuals(model))
qqline(residuals(model), col = "red")
```

```r


dependent_variable <- df$Price
hist(dependent_variable,
     main = "Histogram of Price",
     xlab = "Price",
     col = "skyblue",
     border = "white")

#Homoscedasticity
# Get the standardized residuals from the model
standardized_residuals <- rstandard(model)

# Get the fitted (predicted) values from the model
fitted_values <- fitted(model)

# Plot the standardized residuals against the fitted values
plot(fitted_values, standardized_residuals,
     main = "Standardized Residuals vs Fitted Values",
     xlab = "Fitted (Predicted) Values",
     ylab = "Standardized Residuals")

# Add a horizontal line at y=0 for reference
abline(h = 0, col = "red", lty = 2)

#full model
model <- lm(Price~Age+KM+Color+CC+Weight, data = df)
anova(model)
#forward_method

ols_step_forward_p(model,pent=0.05,details=T)

forward_ml<-ols_step_forward_p(model,pent=0.05,details=F)
forward_ml

```

```
#reduced model
model_2<-lm(Price~Age+KM+Color+Weight,data=df)
model_2
summary(model_2)
#VIF Values of the model
vif(model)

# Define the parameters
alpha <- 0.05        # The significance level (alpha)
v1 <- 3             # Numerator degrees of freedom
v2 <- 235            # Denominator degrees of freedom

# Calculate the critical F-value using the qf() function
critical_F_value <- qf(1 - alpha, v1, v2)

# Print the result
print(critical_F_value)
```